# GWO — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

## Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

## Resumen — posicion entre los 11 algoritmos

| dimension | presupuesto | rango medio de Friedman | posicion de 11 | funciones donde es el mejor |
|---|---|---|---|---|
| D10 | 5k FES | 3.68 | 4º | 7 de 71 |
| D10 | 50k FES | 5.38 | 6º | 3 de 71 |
| D10 | 500k FES | 6.87 | 7º | 1 de 71 |
| D50 | 5k FES | 2.51 | 2º | 13 de 71 |
| D50 | 50k FES | 4.55 | 4º | 7 de 71 |
| D50 | 500k FES | 7.03 | 8º | 1 de 71 |

# GWO — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

| func | suite | dim | 5k | | | | | 50k | | | | | 500k | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | b | w | med | avg | sd | b | w | med | avg | sd | b | w | med | avg | sd |
| F12014 | 2014 | 10 | 7.75e+05 | 4.45e+07 | 1.09e+07 | 1.22e+07 | 8.71e+06 | 4.89e+05 | 3.85e+07 | 4.46e+06 | 6.56e+06 | 6.10e+06 | 1.82e+05 | 1.08e+07 | 3.54e+06 | 3.63e+06 | 2.75e+06 |
| F22014 | 2014 | 10 | 6.28e+04 | 2.82e+08 | 3.35e+05 | 8.54e+06 | 4.25e+07 | 7.13e+02 | 2.33e+04 | 3.70e+03 | 5.81e+03 | 5.01e+03 | 9.63e+00 | 1.19e+04 | 8.00e+02 | 3.33e+03 | 4.23e+03 |
| F32014 | 2014 | 10 | 2.19e+03 | 5.78e+04 | 2.03e+04 | 2.05e+04 | 1.29e+04 | 2.43e+02 | 1.79e+04 | 6.00e+03 | 7.72e+03 | 5.08e+03 | 1.87e+02 | 1.37e+04 | 1.62e+03 | 3.17e+03 | 3.20e+03 |
| F42014 | 2014 | 10 | 6.26e+00 | 5.56e+01 | 3.59e+01 | 3.44e+01 | 8.31e+00 | 4.99e+00 | 3.86e+01 | 3.53e+01 | 3.23e+01 | 8.99e+00 | 2.94e-01 | 3.79e+01 | 3.50e+01 | 2.88e+01 | 1.15e+01 |
| F52014 | 2014 | 10 | 2.03e+01 | 2.08e+01 | 2.06e+01 | 2.06e+01 | 1.14e-01 | 4.22e+00 | 2.05e+01 | 2.04e+01 | 2.01e+01 | 2.27e+00 | 3.84e-01 | 2.04e+01 | 2.03e+01 | 1.99e+01 | 2.79e+00 |
| F62014 | 2014 | 10 | 7.82e-01 | 5.98e+00 | 2.51e+00 | 2.61e+00 | 1.16e+00 | 3.43e-01 | 5.74e+00 | 1.69e+00 | 1.99e+00 | 1.13e+00 | 7.91e-02 | 5.02e+00 | 1.14e+00 | 1.42e+00 | 1.06e+00 |
| F72014 | 2014 | 10 | 7.64e-01 | 2.62e+00 | 1.10e+00 | 1.35e+00 | 5.21e-01 | 1.41e-01 | 3.66e+00 | 8.73e-01 | 1.11e+00 | 7.59e-01 | 3.56e-02 | 3.70e+00 | 5.73e-01 | 8.66e-01 | 8.83e-01 |
| F82014 | 2014 | 10 | 7.31e+00 | 5.77e+01 | 2.12e+01 | 2.58e+01 | 1.51e+01 | 1.99e+00 | 3.76e+01 | 1.09e+01 | 1.25e+01 | 7.44e+00 | 9.95e-01 | 2.64e+01 | 7.96e+00 | 8.97e+00 | 4.64e+00 |
| F92014 | 2014 | 10 | 5.59e+00 | 5.54e+01 | 2.06e+01 | 2.50e+01 | 1.38e+01 | 4.12e+00 | 3.79e+01 | 1.53e+01 | 1.73e+01 | 8.87e+00 | 3.11e+00 | 3.62e+01 | 1.10e+01 | 1.27e+01 | 6.23e+00 |
| F102014 | 2014 | 10 | 1.61e+02 | 2.11e+03 | 6.89e+02 | 8.52e+02 | 5.26e+02 | 1.72e+01 | 1.46e+03 | 3.87e+02 | 4.69e+02 | 3.33e+02 | 7.59e+00 | 8.02e+02 | 2.57e+02 | 2.58e+02 | 1.74e+02 |
| F112014 | 2014 | 10 | 1.68e+01 | 2.23e+03 | 8.77e+02 | 1.05e+03 | 5.94e+02 | 2.73e+01 | 1.81e+03 | 5.21e+02 | 6.76e+02 | 4.75e+02 | 4.00e+00 | 1.41e+03 | 3.60e+02 | 4.09e+02 | 2.61e+02 |
| F122014 | 2014 | 10 | 1.57e-01 | 2.93e+00 | 1.92e+00 | 1.88e+00 | 5.90e-01 | 3.99e-02 | 1.75e+00 | 1.24e+00 | 1.13e+00 | 4.32e-01 | 3.74e-03 | 1.23e+00 | 8.55e-01 | 6.73e-01 | 3.93e-01 |
| F132014 | 2014 | 10 | 2.02e-01 | 5.55e-01 | 3.27e-01 | 3.48e-01 | 8.98e-02 | 6.53e-02 | 3.80e-01 | 1.94e-01 | 1.98e-01 | 6.17e-02 | 5.92e-02 | 2.91e-01 | 1.57e-01 | 1.56e-01 | 5.97e-02 |
| F142014 | 2014 | 10 | 2.29e-01 | 7.84e-01 | 3.66e-01 | 4.27e-01 | 1.57e-01 | 8.30e-02 | 6.83e-01 | 2.50e-01 | 3.36e-01 | 1.95e-01 | 1.92e-02 | 6.47e-01 | 1.55e-01 | 2.10e-01 | 1.75e-01 |
| F152014 | 2014 | 10 | 1.51e+00 | 5.21e+00 | 3.55e+00 | 3.39e+00 | 8.89e-01 | 5.53e-01 | 3.84e+00 | 1.61e+00 | 1.80e+00 | 8.74e-01 | 3.26e-01 | 2.60e+00 | 1.26e+00 | 1.27e+00 | 5.38e-01 |
| F162014 | 2014 | 10 | 2.43e+00 | 4.06e+00 | 3.13e+00 | 3.16e+00 | 3.24e-01 | 2.03e+00 | 3.54e+00 | 2.76e+00 | 2.74e+00 | 3.75e-01 | 1.24e+00 | 3.22e+00 | 2.28e+00 | 2.31e+00 | 4.07e-01 |
| F172014 | 2014 | 10 | 2.66e+03 | 2.48e+05 | 2.53e+04 | 5.20e+04 | 6.08e+04 | 1.67e+03 | 9.73e+04 | 1.49e+04 | 1.85e+04 | 1.63e+04 | 2.42e+03 | 2.15e+04 | 9.34e+03 | 1.00e+04 | 4.97e+03 |
| F182014 | 2014 | 10 | 1.00e+04 | 6.60e+04 | 3.64e+04 | 3.81e+04 | 1.38e+04 | 1.05e+04 | 5.49e+04 | 2.82e+04 | 2.99e+04 | 1.12e+04 | 5.90e+03 | 3.50e+04 | 2.01e+04 | 2.03e+04 | 7.66e+03 |
| F192014 | 2014 | 10 | 5.99e+00 | 5.29e+02 | 1.52e+01 | 3.16e+01 | 7.43e+01 | 3.00e+00 | 8.33e+01 | 9.83e+00 | 1.47e+01 | 1.49e+01 | 2.08e+00 | 5.16e+01 | 6.50e+00 | 8.34e+00 | 7.11e+00 |
| F202014 | 2014 | 10 | 7.78e+03 | 7.07e+04 | 2.43e+04 | 2.76e+04 | 1.49e+04 | 3.81e+03 | 7.39e+04 | 2.16e+04 | 2.36e+04 | 1.21e+04 | 5.51e+03 | 4.51e+04 | 1.57e+04 | 1.75e+04 | 8.43e+03 |
| F212014 | 2014 | 10 | 9.33e+02 | 6.85e+05 | 1.09e+04 | 2.82e+04 | 9.44e+04 | 1.19e+03 | 5.59e+05 | 7.94e+03 | 1.99e+04 | 7.73e+04 | 9.28e+02 | 1.09e+04 | 5.08e+03 | 5.07e+03 | 2.50e+03 |
| F222014 | 2014 | 10 | 2.36e+01 | 7.42e+02 | 2.42e+02 | 1.97e+02 | 1.61e+02 | 2.04e+01 | 3.74e+02 | 2.39e+02 | 1.64e+02 | 1.20e+02 | 2.02e+01 | 2.44e+02 | 2.37e+02 | 1.35e+02 | 1.08e+02 |
| F232014 | 2014 | 10 | 2.00e+02 | 2.54e+02 | 2.42e+02 | 2.41e+02 | 6.29e+00 | 2.00e+02 | 2.42e+02 | 2.00e+02 | 2.02e+02 | 9.22e+00 | 2.00e+02 | 2.37e+02 | 2.00e+02 | 2.01e+02 | 7.22e+00 |
| F242014 | 2014 | 10 | 1.13e+02 | 1.61e+02 | 1.23e+02 | 1.27e+02 | 1.30e+01 | 1.10e+02 | 1.46e+02 | 1.21e+02 | 1.21e+02 | 7.77e+00 | 1.09e+02 | 1.38e+02 | 1.17e+02 | 1.18e+02 | 4.85e+00 |
| F252014 | 2014 | 10 | 2.00e+02 | 2.04e+02 | 2.02e+02 | 2.02e+02 | 8.81e-01 | 1.75e+02 | 2.04e+02 | 2.01e+02 | 2.01e+02 | 3.71e+00 | 2.00e+02 | 2.02e+02 | 2.01e+02 | 2.01e+02 | 4.24e-01 |
| F262014 | 2014 | 10 | 1.00e+02 | 1.01e+02 | 1.00e+02 | 1.00e+02 | 9.29e-02 | 1.00e+02 | 1.00e+02 | 1.00e+02 | 1.00e+02 | 7.85e-02 | 1.00e+02 | 1.00e+02 | 1.00e+02 | 1.00e+02 | 7.39e-02 |
| F272014 | 2014 | 10 | 5.66e+00 | 4.56e+02 | 4.02e+02 | 3.83e+02 | 5.93e+01 | 6.02e+00 | 4.33e+02 | 4.01e+02 | 3.79e+02 | 6.28e+01 | 5.71e+00 | 4.04e+02 | 4.01e+02 | 3.73e+02 | 6.25e+01 |
| F282014 | 2014 | 10 | 3.70e+02 | 1.05e+03 | 4.60e+02 | 4.80e+02 | 1.17e+02 | 1.41e+02 | 6.85e+02 | 4.28e+02 | 4.39e+02 | 8.89e+01 | 2.53e+02 | 6.56e+02 | 3.79e+02 | 4.28e+02 | 7.76e+01 |
| F292014 | 2014 | 10 | 4.51e+06 | 1.18e+07 | 6.15e+06 | 6.58e+06 | 1.55e+06 | 4.50e+06 | 8.49e+06 | 5.31e+06 | 5.71e+06 | 9.49e+05 | 4.50e+06 | 7.53e+06 | 5.31e+06 | 5.44e+06 | 7.19e+05 |
| F302014 | 2014 | 10 | 6.73e+03 | 9.40e+07 | 3.29e+05 | 6.63e+06 | 1.86e+07 | 4.00e+03 | 2.68e+07 | 1.62e+05 | 2.18e+06 | 5.93e+06 | 3.77e+03 | 2.60e+07 | 8.57e+04 | 9.61e+05 | 3.87e+06 |
| F12017 | 2017 | 10 | 2.66e+05 | 4.96e+08 | 2.24e+06 | 3.13e+07 | 9.38e+07 | 3.08e+03 | 5.01e+08 | 3.96e+04 | 1.90e+07 | 8.64e+07 | 1.22e+02 | 3.48e+08 | 6.04e+03 | 8.63e+06 | 4.90e+07 |
| F22017 | 2017 | 10 | 1.56e+00 | 4.53e+03 | 9.96e+01 | 4.97e+02 | 8.12e+02 | 3.16e-01 | 4.00e+02 | 1.09e+01 | 4.90e+01 | 8.89e+01 | 5.04e-03 | 1.72e+03 | 3.21e+00 | 7.49e+01 | 2.50e+02 |
| F32017 | 2017 | 10 | 3.63e+00 | 1.04e+02 | 1.65e+01 | 2.59e+01 | 2.32e+01 | 3.34e+00 | 6.96e+01 | 2.35e+01 | 2.09e+01 | 1.45e+01 | 1.49e-02 | 3.26e+01 | 9.03e+00 | 1.60e+01 | 1.12e+01 |
| F42017 | 2017 | 10 | 3.68e+01 | 4.06e+02 | 7.13e+01 | 9.54e+01 | 6.37e+01 | 8.23e+00 | 3.57e+02 | 4.86e+01 | 6.02e+01 | 5.10e+01 | 4.17e+00 | 4.04e+02 | 2.83e+01 | 4.93e+01 | 7.44e+01 |
| F52017 | 2017 | 10 | 1.22e-08 | 8.90e-03 | 2.90e-05 | 8.39e-04 | 1.77e-03 | 4.13e-11 | 2.11e-03 | 8.74e-07 | 9.24e-05 | 3.28e-04 | 0.00e+00 | 1.27e-04 | 1.71e-08 | 8.76e-06 | 2.64e-05 |
| F62017 | 2017 | 10 | 8.06e+01 | 7.39e+03 | 2.66e+02 | 9.13e+02 | 1.34e+03 | 3.28e+01 | 6.47e+03 | 1.00e+02 | 5.15e+02 | 1.02e+03 | 7.57e+00 | 5.91e+03 | 4.82e+01 | 5.06e+02 | 9.78e+02 |
| F72017 | 2017 | 10 | 1.40e+01 | 9.60e+01 | 4.51e+01 | 4.60e+01 | 1.84e+01 | 1.20e+01 | 5.07e+01 | 2.60e+01 | 2.79e+01 | 8.69e+00 | 1.02e+01 | 3.60e+01 | 2.20e+01 | 2.27e+01 | 6.36e+00 |
| F82017 | 2017 | 10 | 3.50e-04 | 1.03e+00 | 8.93e-02 | 1.22e-01 | 1.96e-01 | 5.17e-05 | 1.16e+00 | 2.23e-02 | 8.19e-02 | 1.87e-01 | 8.17e-08 | 1.16e+00 | 9.06e-02 | 1.04e-01 | 1.87e-01 |
| F92017 | 2017 | 10 | 1.52e+01 | 1.91e+03 | 7.52e+02 | 9.48e+02 | 5.37e+02 | 4.02e+01 | 1.53e+03 | 3.68e+02 | 4.34e+02 | 3.13e+02 | 7.61e+00 | 6.42e+02 | 1.70e+02 | 2.48e+02 | 1.88e+02 |
| F102017 | 2017 | 10 | 3.09e+03 | 5.16e+04 | 1.61e+04 | 1.71e+04 | 1.02e+04 | 2.61e+03 | 3.60e+04 | 1.11e+04 | 1.30e+04 | 7.58e+03 | 4.54e+02 | 3.09e+04 | 7.47e+03 | 8.31e+03 | 5.76e+03 |
| F112017 | 2017 | 10 | 1.51e+04 | 2.77e+07 | 2.97e+06 | 5.35e+06 | 6.11e+06 | 4.05e+03 | 1.25e+07 | 2.66e+06 | 2.99e+06 | 2.89e+06 | 3.26e+03 | 6.41e+06 | 2.41e+06 | 2.25e+06 | 1.85e+06 |
| F122017 | 2017 | 10 | 3.71e+03 | 3.69e+05 | 1.84e+04 | 7.00e+04 | 1.05e+05 | 3.87e+03 | 1.64e+05 | 1.39e+04 | 3.06e+04 | 3.64e+04 | 4.04e+03 | 5.48e+04 | 1.10e+04 | 1.32e+04 | 9.04e+03 |
| F132017 | 2017 | 10 | 5.45e+03 | 4.02e+04 | 1.48e+04 | 1.71e+04 | 8.83e+03 | 1.98e+02 | 2.28e+04 | 6.97e+03 | 7.98e+03 | 5.17e+03 | 3.41e+01 | 1.32e+04 | 1.14e+03 | 3.25e+03 | 3.37e+03 |
| F142017 | 2017 | 10 | 7.06e+03 | 9.35e+05 | 1.36e+05 | 1.93e+05 | 2.09e+05 | 8.87e+03 | 3.23e+05 | 2.65e+04 | 8.15e+04 | 8.80e+04 | 5.47e+03 | 2.89e+05 | 1.52e+04 | 5.37e+04 | 7.45e+04 |
| F152017 | 2017 | 10 | 1.63e+02 | 1.98e+04 | 1.47e+03 | 2.82e+03 | 4.22e+03 | 4.12e+00 | 2.22e+03 | 7.77e+02 | 7.82e+02 | 5.42e+02 | 3.07e+00 | 1.49e+03 | 4.03e+02 | 4.99e+02 | 4.44e+02 |
| F162017 | 2017 | 10 | 5.99e+01 | 3.87e+04 | 4.10e+03 | 8.83e+03 | 1.07e+04 | 3.39e+01 | 2.05e+04 | 8.93e+02 | 1.93e+03 | 3.23e+03 | 2.25e+01 | 4.81e+03 | 4.81e+02 | 5.56e+02 | 7.10e+02 |
| F172017 | 2017 | 10 | 5.96e+03 | 5.59e+04 | 2.49e+04 | 2.71e+04 | 1.11e+04 | 3.19e+03 | 3.91e+04 | 2.08e+04 | 1.93e+04 | 8.70e+03 | 3.26e+03 | 3.29e+04 | 1.21e+04 | 1.30e+04 | 7.66e+03 |
| F182017 | 2017 | 10 | 2.56e+02 | 3.84e+04 | 1.08e+04 | 1.41e+04 | 9.59e+03 | 1.22e+03 | 3.34e+04 | 9.18e+03 | 1.09e+04 | 7.79e+03 | 2.29e+01 | 1.59e+04 | 3.19e+03 | 5.28e+03 | 4.87e+03 |
| F192017 | 2017 | 10 | 3.50e+01 | 4.09e+02 | 7.74e+01 | 1.12e+02 | 8.44e+01 | 1.96e+01 | 2.64e+02 | 5.80e+01 | 8.25e+01 | 6.44e+01 | 1.30e+01 | 1.80e+02 | 3.87e+01 | 5.20e+01 | 3.55e+01 |

| func | suite | dim | 5k | | | | | 50k | | | | | 500k | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | b | w | med | avg | sd | b | w | med | avg | sd | b | w | med | avg | sd |
| F202017 | 2017 | 10 | 1.40e+02 | 2.79e+02 | 2.23e+02 | 2.19e+02 | 3.50e+01 | 1.04e+02 | 2.85e+02 | 2.02e+02 | 1.94e+02 | 3.86e+01 | 1.01e+02 | 2.66e+02 | 1.87e+02 | 1.76e+02 | 4.14e+01 |
| F212017 | 2017 | 10 | 1.01e+02 | 1.11e+02 | 1.05e+02 | 1.05e+02 | 2.72e+00 | 1.00e+02 | 1.10e+02 | 1.04e+02 | 1.05e+02 | 2.92e+00 | 1.00e+02 | 1.10e+02 | 1.04e+02 | 1.04e+02 | 2.27e+00 |
| F222017 | 2017 | 10 | 1.24e+02 | 5.25e+02 | 2.41e+02 | 2.55e+02 | 9.46e+01 | 1.02e+02 | 5.99e+02 | 2.03e+02 | 2.11e+02 | 8.99e+01 | 1.00e+02 | 5.73e+02 | 2.00e+02 | 2.00e+02 | 8.82e+01 |
| F232017 | 2017 | 10 | 2.34e+02 | 8.92e+02 | 4.29e+02 | 4.84e+02 | 1.66e+02 | 2.10e+02 | 6.74e+02 | 2.76e+02 | 3.58e+02 | 1.67e+02 | 2.06e+02 | 6.56e+02 | 2.48e+02 | 3.42e+02 | 1.79e+02 |
| F242017 | 2017 | 10 | 4.79e+02 | 5.86e+02 | 4.91e+02 | 4.99e+02 | 2.26e+01 | 4.77e+02 | 5.19e+02 | 4.86e+02 | 4.91e+02 | 1.19e+01 | 4.76e+02 | 5.18e+02 | 4.84e+02 | 4.90e+02 | 1.25e+01 |
| F252017 | 2017 | 10 | 4.58e+02 | 4.89e+02 | 4.75e+02 | 4.74e+02 | 7.03e+00 | 4.58e+02 | 4.86e+02 | 4.67e+02 | 4.67e+02 | 5.95e+00 | 4.57e+02 | 4.86e+02 | 4.67e+02 | 4.67e+02 | 4.89e+00 |
| F262017 | 2017 | 10 | 3.91e+02 | 4.80e+02 | 3.96e+02 | 4.03e+02 | 1.89e+01 | 3.90e+02 | 4.76e+02 | 3.94e+02 | 3.95e+02 | 1.22e+01 | 3.88e+02 | 4.34e+02 | 3.91e+02 | 3.94e+02 | 8.04e+00 |
| F272017 | 2017 | 10 | 2.60e+01 | 4.77e+02 | 1.19e+02 | 1.71e+02 | 1.55e+02 | 9.18e-01 | 4.37e+02 | 1.05e+01 | 9.86e+01 | 1.39e+02 | 5.79e-02 | 4.38e+02 | 2.21e-01 | 5.77e+01 | 1.24e+02 |
| F282017 | 2017 | 10 | 1.62e+03 | 1.59e+06 | 1.93e+04 | 6.30e+04 | 2.21e+05 | 1.51e+03 | 6.57e+04 | 8.06e+03 | 1.32e+04 | 1.34e+04 | 1.45e+03 | 2.52e+04 | 4.73e+03 | 5.97e+03 | 5.42e+03 |
| F292017 | 2017 | 10 | 1.39e+04 | 4.79e+07 | 5.34e+04 | 1.44e+06 | 6.91e+06 | 3.13e+03 | 4.64e+05 | 2.79e+04 | 5.17e+04 | 7.46e+04 | 4.08e+03 | 1.47e+05 | 2.56e+04 | 3.07e+04 | 2.70e+04 |
| F12022 | 2022 | 10 | 2.86e+00 | 1.69e+03 | 8.63e+01 | 1.52e+02 | 2.96e+02 | 5.00e+00 | 2.19e+02 | 6.39e+01 | 7.09e+01 | 4.77e+01 | 4.47e-03 | 1.52e+02 | 4.24e+01 | 4.36e+01 | 3.69e+01 |
| F22022 | 2022 | 10 | 6.18e+00 | 7.75e+01 | 1.32e+01 | 2.78e+01 | 2.29e+01 | 4.88e-01 | 7.18e+01 | 1.12e+01 | 2.16e+01 | 2.07e+01 | 1.25e-01 | 7.23e+01 | 1.08e+01 | 1.72e+01 | 1.97e+01 |
| F32022 | 2022 | 10 | 8.50e-06 | 2.63e-02 | 2.74e-04 | 1.33e-03 | 3.78e-03 | 5.08e-08 | 1.32e-02 | 8.40e-05 | 7.63e-04 | 1.92e-03 | 3.36e-09 | 1.36e-02 | 1.47e-06 | 7.64e-04 | 1.96e-03 |
| F42022 | 2022 | 10 | 1.40e+01 | 1.06e+02 | 4.56e+01 | 4.88e+01 | 1.98e+01 | 1.20e+01 | 5.85e+01 | 3.00e+01 | 3.00e+01 | 1.07e+01 | 1.00e+01 | 3.60e+01 | 2.12e+01 | 2.18e+01 | 6.33e+00 |
| F52022 | 2022 | 10 | 1.24e-04 | 5.37e-01 | 1.60e-02 | 7.91e-02 | 1.35e-01 | 1.39e-04 | 5.36e-01 | 5.31e-03 | 5.76e-02 | 1.37e-01 | 4.73e-06 | 5.64e-01 | 2.60e-03 | 5.45e-02 | 1.54e-01 |
| F62022 | 2022 | 10 | 7.82e+03 | 9.36e+04 | 3.41e+04 | 3.82e+04 | 1.87e+04 | 7.45e+03 | 7.53e+04 | 3.26e+04 | 3.44e+04 | 1.66e+04 | 7.35e+03 | 6.34e+04 | 3.11e+04 | 3.41e+04 | 1.66e+04 |
| F72022 | 2022 | 10 | 2.63e+01 | 1.98e+02 | 4.52e+01 | 6.85e+01 | 4.59e+01 | 2.28e+01 | 1.41e+02 | 3.16e+01 | 4.16e+01 | 3.08e+01 | 1.04e+01 | 1.40e+02 | 2.49e+01 | 2.93e+01 | 2.16e+01 |
| F82022 | 2022 | 10 | 3.03e+01 | 2.95e+03 | 5.61e+02 | 8.09e+02 | 7.72e+02 | 2.27e+01 | 1.85e+03 | 4.73e+01 | 2.34e+02 | 5.15e+02 | 2.64e+00 | 1.37e+02 | 2.44e+01 | 2.94e+01 | 1.89e+01 |
| F92022 | 2022 | 10 | 4.64e+00 | 4.99e+02 | 3.74e+02 | 3.10e+02 | 1.71e+02 | 3.12e-01 | 4.89e+02 | 3.66e+02 | 2.79e+02 | 1.73e+02 | 2.31e-02 | 4.15e+02 | 2.06e+02 | 1.86e+02 | 1.84e+02 |
| F102022 | 2022 | 10 | 2.21e+02 | 3.69e+02 | 2.24e+02 | 2.52e+02 | 4.92e+01 | 2.21e+02 | 3.59e+02 | 2.23e+02 | 2.51e+02 | 4.82e+01 | 2.21e+02 | 3.34e+02 | 2.22e+02 | 2.49e+02 | 4.44e+01 |
| F112022 | 2022 | 10 | 1.99e-01 | 8.79e+02 | 3.44e+00 | 3.85e+01 | 1.71e+02 | 7.60e-02 | 8.77e+02 | 3.20e+00 | 3.78e+01 | 1.71e+02 | 2.93e-03 | 8.76e+02 | 3.60e-02 | 3.61e+01 | 1.71e+02 |
| F122022 | 2022 | 10 | 1.64e+02 | 2.13e+02 | 1.67e+02 | 1.70e+02 | 9.22e+00 | 1.64e+02 | 1.82e+02 | 1.65e+02 | 1.66e+02 | 3.07e+00 | 1.64e+02 | 1.81e+02 | 1.65e+02 | 1.66e+02 | 2.75e+00 |

# GWO — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

| func | suite | dim | 5k | | | | | 50k | | | | | 500k | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | b | w | med | avg | sd | b | w | med | avg | sd | b | w | med | avg | sd |
| F12014 | 2014 | 50 | 7.48e+07 | 4.06e+08 | 2.04e+08 | 2.02e+08 | 7.07e+07 | 3.20e+07 | 2.89e+08 | 9.69e+07 | 1.07e+08 | 5.37e+07 | 2.49e+07 | 1.94e+08 | 6.37e+07 | 7.45e+07 | 3.58e+07 |
| F22014 | 2014 | 50 | 3.07e+09 | 1.65e+10 | 8.40e+09 | 8.38e+09 | 3.18e+09 | 9.05e+08 | 1.46e+10 | 6.73e+09 | 6.97e+09 | 2.75e+09 | 7.83e+08 | 1.55e+10 | 6.32e+09 | 6.35e+09 | 2.92e+09 |
| F32014 | 2014 | 50 | 1.46e+05 | 3.46e+05 | 2.11e+05 | 2.18e+05 | 4.76e+04 | 4.33e+04 | 1.61e+05 | 9.79e+04 | 1.01e+05 | 2.25e+04 | 2.65e+04 | 9.03e+04 | 5.81e+04 | 5.89e+04 | 1.32e+04 |
| F42014 | 2014 | 50 | 3.89e+02 | 3.00e+03 | 1.09e+03 | 1.20e+03 | 5.51e+02 | 2.34e+02 | 1.94e+03 | 6.36e+02 | 7.15e+02 | 3.42e+02 | 2.41e+02 | 1.38e+03 | 5.36e+02 | 5.87e+02 | 2.48e+02 |
| F52014 | 2014 | 50 | 2.12e+01 | 2.14e+01 | 2.13e+01 | 2.13e+01 | 4.20e-02 | 2.11e+01 | 2.13e+01 | 2.12e+01 | 2.12e+01 | 3.07e-02 | 2.11e+01 | 2.12e+01 | 2.11e+01 | 2.11e+01 | 2.68e-02 |
| F62014 | 2014 | 50 | 2.98e+01 | 4.97e+01 | 3.89e+01 | 3.89e+01 | 4.08e+00 | 2.48e+01 | 4.36e+01 | 3.37e+01 | 3.36e+01 | 4.47e+00 | 2.21e+01 | 3.95e+01 | 2.98e+01 | 3.04e+01 | 4.33e+00 |
| F72014 | 2014 | 50 | 2.16e+01 | 1.80e+02 | 8.09e+01 | 8.31e+01 | 3.69e+01 | 1.70e+01 | 1.72e+02 | 4.72e+01 | 5.45e+01 | 2.83e+01 | 1.24e+01 | 1.03e+02 | 4.08e+01 | 4.51e+01 | 2.40e+01 |
| F82014 | 2014 | 50 | 1.55e+02 | 5.02e+02 | 2.51e+02 | 2.75e+02 | 8.75e+01 | 1.25e+02 | 3.98e+02 | 1.88e+02 | 1.97e+02 | 5.82e+01 | 1.16e+02 | 3.83e+02 | 1.85e+02 | 1.87e+02 | 3.89e+01 |
| F92014 | 2014 | 50 | 1.88e+02 | 5.42e+02 | 2.84e+02 | 3.09e+02 | 8.39e+01 | 1.64e+02 | 5.35e+02 | 2.14e+02 | 2.32e+02 | 7.95e+01 | 1.33e+02 | 4.46e+02 | 1.83e+02 | 1.96e+02 | 5.82e+01 |
| F102014 | 2014 | 50 | 5.67e+03 | 1.57e+04 | 8.05e+03 | 9.20e+03 | 2.99e+03 | 4.22e+03 | 1.42e+04 | 5.92e+03 | 6.52e+03 | 2.14e+03 | 3.64e+03 | 1.14e+04 | 5.55e+03 | 5.71e+03 | 1.31e+03 |
| F112014 | 2014 | 50 | 6.76e+03 | 1.57e+04 | 1.45e+04 | 1.27e+04 | 2.93e+03 | 3.33e+03 | 1.46e+04 | 7.72e+03 | 9.65e+03 | 3.89e+03 | 3.61e+03 | 1.37e+04 | 5.80e+03 | 6.38e+03 | 2.09e+03 |
| F122014 | 2014 | 50 | 3.50e+00 | 5.72e+00 | 4.85e+00 | 4.75e+00 | 4.92e-01 | 2.77e-01 | 4.65e+00 | 4.05e+00 | 3.77e+00 | 1.01e+00 | 6.86e-02 | 3.85e+00 | 3.35e+00 | 3.01e+00 | 1.09e+00 |
| F132014 | 2014 | 50 | 6.37e-01 | 2.78e+00 | 8.72e-01 | 9.96e-01 | 4.67e-01 | 5.63e-01 | 2.23e+00 | 7.74e-01 | 8.38e-01 | 3.32e-01 | 4.97e-01 | 8.54e-01 | 6.44e-01 | 6.53e-01 | 7.46e-02 |
| F142014 | 2014 | 50 | 4.41e-01 | 5.60e+01 | 1.93e+01 | 1.87e+01 | 1.30e+01 | 3.06e-01 | 3.12e+01 | 1.00e+01 | 1.17e+01 | 9.98e+00 | 2.37e-01 | 3.55e+01 | 4.36e+00 | 1.06e+01 | 1.15e+01 |
| F152014 | 2014 | 50 | 7.39e+01 | 2.19e+04 | 2.17e+03 | 3.70e+03 | 4.90e+03 | 5.83e+01 | 5.77e+03 | 4.38e+02 | 9.44e+02 | 1.26e+03 | 2.34e+01 | 2.17e+04 | 2.50e+02 | 1.51e+03 | 3.76e+03 |
| F162014 | 2014 | 50 | 2.16e+01 | 2.38e+01 | 2.28e+01 | 2.28e+01 | 4.95e-01 | 1.95e+01 | 2.34e+01 | 2.19e+01 | 2.18e+01 | 7.56e-01 | 1.88e+01 | 2.25e+01 | 2.12e+01 | 2.10e+01 | 7.42e-01 |
| F172014 | 2014 | 50 | 1.45e+06 | 4.85e+07 | 1.29e+07 | 1.44e+07 | 1.03e+07 | 7.80e+05 | 2.78e+07 | 3.52e+06 | 4.93e+06 | 4.41e+06 | 3.24e+05 | 1.04e+07 | 1.76e+06 | 2.41e+06 | 2.04e+06 |
| F182014 | 2014 | 50 | 3.43e+05 | 9.60e+07 | 1.83e+06 | 4.90e+06 | 1.34e+07 | 1.10e+05 | 2.31e+06 | 1.90e+05 | 3.45e+05 | 4.31e+05 | 1.01e+05 | 1.77e+07 | 1.63e+05 | 6.30e+05 | 2.49e+06 |
| F192014 | 2014 | 50 | 3.59e+02 | 6.09e+05 | 8.21e+03 | 5.22e+04 | 1.06e+05 | 6.16e+01 | 1.54e+05 | 1.25e+03 | 1.16e+04 | 3.01e+04 | 4.05e+01 | 3.58e+04 | 2.37e+02 | 4.09e+03 | 7.38e+03 |
| F202014 | 2014 | 50 | 1.81e+05 | 8.68e+09 | 8.84e+06 | 7.34e+08 | 1.85e+09 | 1.47e+05 | 2.23e+09 | 2.32e+05 | 7.62e+07 | 3.39e+08 | 1.07e+05 | 2.06e+09 | 2.23e+05 | 7.39e+07 | 3.41e+08 |
| F212014 | 2014 | 50 | 6.20e+05 | 1.28e+08 | 1.85e+07 | 2.47e+07 | 2.22e+07 | 7.85e+05 | 4.19e+07 | 4.44e+06 | 8.91e+06 | 1.08e+07 | 2.74e+05 | 4.62e+07 | 1.99e+06 | 3.90e+06 | 6.91e+06 |
| F222014 | 2014 | 50 | 2.97e+03 | 1.44e+06 | 5.25e+03 | 3.85e+04 | 2.03e+05 | 2.74e+03 | 3.55e+06 | 4.20e+03 | 9.08e+04 | 5.09e+05 | 1.14e+03 | 4.86e+03 | 2.97e+03 | 2.97e+03 | 8.32e+02 |
| F232014 | 2014 | 50 | 2.37e+02 | 2.45e+02 | 2.38e+02 | 2.39e+02 | 2.06e+00 | 2.36e+02 | 2.39e+02 | 2.36e+02 | 2.37e+02 | 6.88e-01 | 2.36e+02 | 2.39e+02 | 2.36e+02 | 2.36e+02 | 4.34e-01 |
| F242014 | 2014 | 50 | 2.40e+02 | 3.03e+02 | 2.71e+02 | 2.68e+02 | 1.27e+01 | 2.00e+02 | 2.00e+02 | 2.00e+02 | 2.00e+02 | 3.76e-02 | 2.00e+02 | 2.00e+02 | 2.00e+02 | 2.00e+02 | 3.42e-03 |
| F252014 | 2014 | 50 | 2.36e+02 | 2.66e+02 | 2.49e+02 | 2.49e+02 | 7.45e+00 | 2.00e+02 | 2.48e+02 | 2.37e+02 | 2.34e+02 | 1.07e+01 | 2.00e+02 | 2.44e+02 | 2.30e+02 | 2.31e+02 | 8.11e+00 |
| F262014 | 2014 | 50 | 1.01e+02 | 3.96e+02 | 2.03e+02 | 2.04e+02 | 6.56e+01 | 1.01e+02 | 3.84e+02 | 2.00e+02 | 1.96e+02 | 7.22e+01 | 1.01e+02 | 3.70e+02 | 2.00e+02 | 1.86e+02 | 6.83e+01 |
| F272014 | 2014 | 50 | 1.09e+03 | 1.46e+03 | 1.28e+03 | 1.28e+03 | 8.46e+01 | 7.69e+02 | 1.41e+03 | 1.10e+03 | 1.09e+03 | 1.60e+02 | 7.22e+02 | 1.32e+03 | 1.02e+03 | 1.01e+03 | 1.11e+02 |
| F282014 | 2014 | 50 | 2.32e+03 | 8.04e+03 | 4.15e+03 | 4.25e+03 | 1.08e+03 | 1.59e+03 | 4.32e+03 | 2.45e+03 | 2.63e+03 | 7.22e+02 | 1.47e+03 | 3.39e+03 | 2.15e+03 | 2.20e+03 | 5.17e+02 |
| F292014 | 2014 | 50 | 8.35e+06 | 4.50e+08 | 1.40e+08 | 1.89e+08 | 1.40e+08 | 6.75e+05 | 4.31e+08 | 4.83e+07 | 7.18e+07 | 8.13e+07 | 3.73e+05 | 2.05e+08 | 2.38e+07 | 4.90e+07 | 5.33e+07 |
| F302014 | 2014 | 50 | 1.77e+08 | 2.26e+09 | 8.46e+08 | 9.09e+08 | 4.83e+08 | 2.75e+07 | 1.78e+09 | 3.39e+08 | 3.76e+08 | 2.90e+08 | 1.73e+07 | 7.97e+08 | 2.05e+08 | 2.52e+08 | 1.85e+08 |
| F12017 | 2017 | 50 | 2.89e+09 | 1.75e+10 | 8.48e+09 | 9.04e+09 | 3.74e+09 | 1.64e+09 | 1.41e+10 | 5.60e+09 | 6.10e+09 | 2.87e+09 | 1.83e+09 | 1.04e+10 | 5.09e+09 | 5.13e+09 | 1.84e+09 |
| F22017 | 2017 | 50 | 2.64e+04 | 7.88e+04 | 5.00e+04 | 5.04e+04 | 1.12e+04 | 7.90e+03 | 4.56e+04 | 2.81e+04 | 2.76e+04 | 9.43e+03 | 5.49e+03 | 4.38e+04 | 2.29e+04 | 2.35e+04 | 7.46e+03 |
| F32017 | 2017 | 50 | 2.89e+02 | 2.24e+03 | 7.60e+02 | 8.84e+02 | 4.67e+02 | 2.46e+02 | 1.64e+03 | 3.82e+02 | 4.71e+02 | 2.59e+02 | 1.92e+02 | 1.35e+03 | 3.11e+02 | 3.98e+02 | 2.34e+02 |
| F42017 | 2017 | 50 | 2.77e+03 | 1.74e+04 | 7.11e+03 | 7.92e+03 | 3.61e+03 | 1.58e+03 | 1.28e+04 | 4.47e+03 | 5.57e+03 | 2.99e+03 | 1.64e+03 | 1.43e+04 | 4.47e+03 | 5.05e+03 | 2.48e+03 |
| F52017 | 2017 | 50 | 4.74e-04 | 5.43e-02 | 2.29e-03 | 6.82e-03 | 1.18e-02 | 3.59e-05 | 6.88e-03 | 5.86e-04 | 9.37e-04 | 1.24e-03 | 5.70e-05 | 2.03e-03 | 3.48e-04 | 4.31e-04 | 3.61e-04 |
| F62017 | 2017 | 50 | 9.26e+04 | 6.89e+05 | 3.17e+05 | 3.46e+05 | 1.36e+05 | 5.98e+04 | 6.32e+05 | 2.33e+05 | 2.65e+05 | 1.39e+05 | 3.58e+04 | 4.14e+05 | 2.16e+05 | 2.17e+05 | 8.49e+04 |
| F72017 | 2017 | 50 | 3.25e+02 | 1.04e+03 | 5.88e+02 | 6.47e+02 | 2.16e+02 | 2.76e+02 | 9.23e+02 | 4.01e+02 | 4.45e+02 | 1.53e+02 | 2.46e+02 | 8.56e+02 | 3.67e+02 | 4.14e+02 | 1.49e+02 |
| F82017 | 2017 | 50 | 3.72e+00 | 2.86e+01 | 1.23e+01 | 1.28e+01 | 4.62e+00 | 3.04e+00 | 1.56e+01 | 8.62e+00 | 8.78e+00 | 3.38e+00 | 2.92e+00 | 1.76e+01 | 7.57e+00 | 7.40e+00 | 2.92e+00 |
| F92017 | 2017 | 50 | 7.98e+03 | 1.60e+04 | 1.13e+04 | 1.20e+04 | 2.94e+03 | 4.60e+03 | 1.51e+04 | 7.17e+03 | 9.01e+03 | 3.49e+03 | 4.12e+03 | 1.39e+04 | 6.52e+03 | 7.42e+03 | 2.74e+03 |
| F102017 | 2017 | 50 | 1.68e+05 | 5.15e+07 | 2.91e+06 | 9.49e+06 | 1.37e+07 | 1.14e+05 | 3.31e+07 | 5.66e+05 | 3.12e+06 | 7.26e+06 | 9.01e+04 | 4.27e+07 | 4.93e+05 | 4.03e+06 | 9.38e+06 |
| F112017 | 2017 | 50 | 3.73e+07 | 1.27e+09 | 2.38e+08 | 3.66e+08 | 3.20e+08 | 1.03e+07 | 1.45e+09 | 1.65e+08 | 2.41e+08 | 2.52e+08 | 9.34e+06 | 6.56e+08 | 4.11e+07 | 1.13e+08 | 1.59e+08 |
| F122017 | 2017 | 50 | 8.10e+07 | 3.01e+09 | 4.77e+08 | 6.38e+08 | 5.50e+08 | 8.39e+06 | 1.70e+09 | 2.25e+08 | 3.23e+08 | 3.24e+08 | 2.61e+06 | 2.37e+09 | 1.16e+08 | 2.87e+08 | 3.91e+08 |
| F132017 | 2017 | 50 | 4.94e+05 | 8.44e+06 | 2.52e+06 | 2.91e+06 | 1.86e+06 | 2.67e+05 | 3.95e+06 | 9.07e+05 | 1.11e+06 | 7.39e+05 | 2.21e+05 | 1.73e+06 | 5.00e+05 | 6.24e+05 | 3.63e+05 |
| F142017 | 2017 | 50 | 3.88e+06 | 1.42e+09 | 3.15e+07 | 1.20e+08 | 2.43e+08 | 1.10e+06 | 8.60e+08 | 9.85e+06 | 6.85e+07 | 1.57e+08 | 1.37e+05 | 3.52e+08 | 4.12e+06 | 2.78e+07 | 7.61e+07 |
| F152017 | 2017 | 50 | 5.86e+04 | 5.01e+07 | 1.31e+06 | 5.97e+06 | 1.13e+07 | 3.67e+04 | 4.74e+07 | 6.68e+05 | 3.39e+06 | 9.10e+06 | 3.53e+04 | 2.48e+07 | 1.02e+05 | 2.02e+06 | 4.63e+06 |
| F162017 | 2017 | 50 | 5.09e+04 | 1.12e+05 | 7.26e+04 | 7.48e+04 | 1.63e+04 | 2.09e+04 | 7.58e+04 | 4.88e+04 | 4.81e+04 | 1.13e+04 | 1.44e+04 | 7.75e+04 | 3.10e+04 | 3.16e+04 | 9.60e+03 |
| F172017 | 2017 | 50 | 9.33e+05 | 2.92e+07 | 7.04e+06 | 8.91e+06 | 6.70e+06 | 4.56e+05 | 1.82e+07 | 2.84e+06 | 4.29e+06 | 3.84e+06 | 2.90e+05 | 2.32e+07 | 1.79e+06 | 2.65e+06 | 3.51e+06 |
| F182017 | 2017 | 50 | 8.37e+09 | 8.81e+10 | 2.95e+10 | 3.39e+10 | 1.85e+10 | 6.28e+08 | 5.50e+10 | 1.24e+10 | 1.68e+10 | 1.37e+10 | 1.26e+07 | 3.32e+10 | 8.35e+09 | 1.03e+10 | 8.27e+09 |
| F192017 | 2017 | 50 | 5.81e+02 | 3.34e+03 | 1.80e+03 | 1.83e+03 | 5.92e+02 | 2.46e+02 | 3.14e+03 | 9.57e+02 | 1.10e+03 | 5.80e+02 | 2.27e+02 | 2.01e+03 | 8.27e+02 | 8.62e+02 | 4.46e+02 |

| func | suite | dim | 5k | | | | | 50k | | | | | 500k | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | b | w | med | avg | sd | b | w | med | avg | sd | b | w | med | avg | sd |
| F202017 | 2017 | 50 | 1.39e+03 | 2.28e+04 | 8.22e+03 | 8.45e+03 | 5.11e+03 | 1.34e+03 | 1.57e+04 | 6.12e+03 | 6.31e+03 | 3.51e+03 | 7.39e+02 | 1.99e+04 | 4.79e+03 | 5.30e+03 | 3.56e+03 |
| F212017 | 2017 | 50 | 2.85e+02 | 8.79e+03 | 3.53e+02 | 5.38e+02 | 1.18e+03 | 2.42e+02 | 1.67e+03 | 2.98e+02 | 3.31e+02 | 1.95e+02 | 2.01e+02 | 4.79e+02 | 2.77e+02 | 2.89e+02 | 4.56e+01 |
| F222017 | 2017 | 50 | 5.18e+03 | 3.84e+04 | 1.76e+04 | 1.98e+04 | 8.32e+03 | 5.06e+03 | 2.71e+04 | 1.26e+04 | 1.27e+04 | 4.78e+03 | 4.07e+03 | 1.99e+04 | 1.12e+04 | 1.10e+04 | 3.36e+03 |
| F232017 | 2017 | 50 | 3.40e+03 | 2.92e+04 | 8.67e+03 | 9.63e+03 | 4.30e+03 | 2.58e+03 | 1.28e+04 | 6.63e+03 | 6.73e+03 | 2.03e+03 | 1.02e+03 | 9.87e+03 | 6.22e+03 | 6.29e+03 | 2.05e+03 |
| F242017 | 2017 | 50 | 1.40e+03 | 3.82e+03 | 2.14e+03 | 2.20e+03 | 5.04e+02 | 1.26e+03 | 2.90e+03 | 1.61e+03 | 1.76e+03 | 4.29e+02 | 9.67e+02 | 2.35e+03 | 1.58e+03 | 1.62e+03 | 3.47e+02 |
| F252017 | 2017 | 50 | 1.80e+03 | 3.31e+03 | 2.23e+03 | 2.29e+03 | 3.40e+02 | 1.67e+03 | 2.86e+03 | 1.86e+03 | 1.91e+03 | 2.22e+02 | 1.56e+03 | 2.48e+03 | 1.84e+03 | 1.88e+03 | 1.71e+02 |
| F262017 | 2017 | 50 | 6.85e+02 | 1.15e+03 | 8.52e+02 | 8.64e+02 | 1.11e+02 | 5.93e+02 | 1.03e+03 | 7.12e+02 | 7.35e+02 | 8.60e+01 | 5.88e+02 | 8.76e+02 | 6.70e+02 | 6.77e+02 | 5.81e+01 |
| F272017 | 2017 | 50 | 4.92e+02 | 8.65e+02 | 5.63e+02 | 5.79e+02 | 7.10e+01 | 4.84e+02 | 6.65e+02 | 5.22e+02 | 5.30e+02 | 3.87e+01 | 4.75e+02 | 6.64e+02 | 5.13e+02 | 5.19e+02 | 3.21e+01 |
| F282017 | 2017 | 50 | 1.55e+08 | 5.30e+09 | 1.55e+09 | 1.65e+09 | 1.07e+09 | 3.62e+07 | 1.79e+09 | 3.94e+08 | 5.00e+08 | 3.82e+08 | 2.62e+07 | 1.15e+09 | 2.45e+08 | 2.76e+08 | 2.09e+08 |
| F292017 | 2017 | 50 | 1.13e+09 | 1.80e+10 | 8.61e+09 | 9.05e+09 | 4.14e+09 | 6.63e+08 | 1.24e+10 | 4.25e+09 | 4.88e+09 | 3.04e+09 | 1.97e+08 | 1.19e+10 | 2.18e+09 | 3.06e+09 | 2.49e+09 |
| F12022 | 2022 | 20 | 5.79e+01 | 1.06e+04 | 1.99e+03 | 2.83e+03 | 2.44e+03 | 6.54e+01 | 4.52e+03 | 7.73e+02 | 1.10e+03 | 1.06e+03 | 3.78e+01 | 4.12e+03 | 5.10e+02 | 9.02e+02 | 9.81e+02 |
| F22022 | 2022 | 20 | 5.12e+01 | 2.13e+02 | 9.09e+01 | 1.02e+02 | 3.78e+01 | 4.81e+01 | 1.63e+02 | 7.80e+01 | 8.34e+01 | 2.30e+01 | 4.61e+01 | 1.69e+02 | 7.75e+01 | 8.06e+01 | 2.53e+01 |
| F32022 | 2022 | 20 | 4.13e-04 | 5.74e-02 | 1.74e-02 | 2.01e-02 | 1.39e-02 | 1.66e-05 | 5.31e-02 | 6.42e-03 | 1.11e-02 | 1.31e-02 | 5.75e-07 | 3.80e-02 | 7.30e-03 | 1.11e-02 | 1.12e-02 |
| F42022 | 2022 | 20 | 4.61e+01 | 2.91e+02 | 1.65e+02 | 1.59e+02 | 7.65e+01 | 4.40e+01 | 2.16e+02 | 1.14e+02 | 1.19e+02 | 5.33e+01 | 4.60e+01 | 1.70e+02 | 9.40e+01 | 9.74e+01 | 3.22e+01 |
| F52022 | 2022 | 20 | 5.67e-02 | 3.64e+00 | 7.64e-01 | 1.01e+00 | 8.98e-01 | 1.70e-03 | 3.22e+00 | 4.28e-01 | 6.71e-01 | 6.40e-01 | 8.72e-04 | 1.54e+00 | 6.47e-01 | 5.94e-01 | 3.73e-01 |
| F62022 | 2022 | 20 | 4.27e+04 | 3.42e+07 | 1.82e+05 | 1.57e+06 | 5.11e+06 | 1.56e+04 | 2.37e+07 | 5.99e+04 | 6.22e+05 | 3.36e+06 | 2.40e+04 | 2.19e+06 | 5.77e+04 | 9.93e+04 | 2.99e+05 |
| F72022 | 2022 | 20 | 4.99e+01 | 8.23e+02 | 1.22e+02 | 1.56e+02 | 1.32e+02 | 2.79e+01 | 2.27e+02 | 5.94e+01 | 6.76e+01 | 3.88e+01 | 2.20e+01 | 1.64e+02 | 4.14e+01 | 5.00e+01 | 2.74e+01 |
| F82022 | 2022 | 20 | 1.18e+03 | 1.39e+04 | 4.02e+03 | 4.14e+03 | 2.10e+03 | 7.01e+02 | 5.05e+03 | 1.96e+03 | 2.05e+03 | 9.26e+02 | 3.07e+02 | 3.58e+03 | 1.09e+03 | 1.20e+03 | 6.30e+02 |
| F92022 | 2022 | 20 | 8.94e+01 | 5.32e+02 | 4.17e+02 | 4.18e+02 | 6.22e+01 | 3.40e+02 | 4.77e+02 | 3.72e+02 | 3.84e+02 | 3.84e+01 | 3.38e+02 | 4.48e+02 | 3.70e+02 | 3.77e+02 | 3.02e+01 |
| F102022 | 2022 | 20 | 4.04e+02 | 5.44e+03 | 2.21e+03 | 2.24e+03 | 1.79e+03 | 4.01e+02 | 4.57e+03 | 1.32e+03 | 1.55e+03 | 1.28e+03 | 3.06e+02 | 4.12e+03 | 5.38e+02 | 1.06e+03 | 9.36e+02 |
| F112022 | 2022 | 20 | 6.00e+00 | 1.36e+03 | 2.26e+01 | 6.06e+01 | 2.03e+02 | 3.45e+00 | 2.78e+01 | 1.32e+01 | 1.53e+01 | 5.51e+00 | 6.39e-01 | 4.55e+01 | 1.24e+01 | 1.38e+01 | 7.93e+00 |
| F122022 | 2022 | 20 | 2.44e+02 | 3.55e+02 | 2.92e+02 | 2.92e+02 | 2.70e+01 | 2.43e+02 | 3.19e+02 | 2.62e+02 | 2.66e+02 | 1.71e+01 | 2.38e+02 | 3.05e+02 | 2.58e+02 | 2.60e+02 | 1.53e+01 |